MOHAMED AYMEN AKCHICHE

ML & GENERATIVE AI ENGINEER

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PROFESSIONAL SUMMARY

Computer Science Engineering student specializing in Artificial Intelligence, with hands-on experience building machine learning, NLP, and Generative AI applications. Experienced in developing end-to-end AI systems, including data preprocessing, text representation, semantic retrieval, RAG pipelines, AI agents, tool calling, conversational memory, and interactive deployment. Strong practical experience with Python, Scikit-learn, LangChain, LangGraph, Google Gemini, ChromaDB, FastAPI, Streamlit, NumPy, Pandas, SQL, Git, and NLP tooling. Interested in building reliable, context-aware AI applications that connect language models with structured and unstructured data.

EDUCATION

State Engineer Degree in Artificial Intelligence (5-Year Graduate Program)

Kasdi Merbah University of Ouargla — Currently in 4th Year

TECHNICAL SKILLS

Programming & Data: Python, SQL, NumPy, Pandas, Matplotlib

Machine Learning & AI: Machine Learning, Deep Learning, Scikit-learn

NLP: Natural Language Processing, NLTK, TF-IDF, Text Preprocessing, Sentence Embeddings

Generative AI & LLMs: Generative AI, Google Gemini, Prompt Engineering

RAG & AI Agents: RAG, LangChain, LangGraph, ChromaDB, Vector Embeddings, Tool Calling, Agent Memory, Multi-Agent Systems

APIs & Deployment: FastAPI, Streamlit, REST API Development, Interactive AI Application Deployment

Engineering & Version Control: Git, Data Preprocessing, Feature Engineering, Feature Selection, Missing Value Handling, Outlier Handling, Model Evaluation, Hyperparameter Tuning, Cross-Validation

SELECTED AI & MACHINE LEARNING PROJECTS

AI-Powered E-Learning Agent

Tech: Python, LangChain, LangGraph, Google Gemini, ChromaDB, RAG, Vector Embeddings, Tool Calling, Agent Memory

GitHub: github.com/Akchiche-Mohamed-Aymen/e-learning-chat-agent

- Designed a multi-agent assistant for an English e-learning platform that combines LLM reasoning, structured learning data, specialized tools, RAG, and conversational memory.
- Built a Main Agent to understand user requests, select appropriate tools, combine results, and generate context-aware responses across student, course, exam, instructor, and educational-content use cases.
- Implemented a Retrieval-Augmented Generation pipeline using vector embeddings and ChromaDB to retrieve relevant educational documents and provide grounded answers.
- Implemented a History Agent that retrieves semantically relevant previous interactions and falls back to summarized conversation history when direct retrieval is insufficient, reducing unnecessary context.
- Added practical context and access handling, including role-based access to student information, restricting students to their own data, fuzzy matching for misspelled names, and ambiguity detection.

Poem Genre Classification

Tech: Python, Scikit-learn, Pandas, NLTK, Streamlit, Sentence Embeddings, NLP

GitHub: github.com/Akchiche-Mohamed-Aymen/poem_classification | Demo: poemclassification.streamlit.app/

- Developed a multi-class NLP system that classifies poems into Affection, Death, Environment, and Music.
- Prepared and cleaned textual data, then fine-tuned a pre-trained sentence embedding model using similarity-based learning to produce domain-specific semantic representations.
- Used the resulting embeddings for downstream classification and evaluated the system using Accuracy, Precision, Recall, and F1-Score.
- Deployed the application as an interactive Streamlit web app.

Spam Classification

Tech: Python, Scikit-learn, NLTK, Pandas, Streamlit, TF-IDF, NLP

GitHub: github.com/Akchiche-Mohamed-Aymen/spam_detection_classifier | Demo: spamdetectionclassifier.streamlit.app/

- Built a binary NLP application to classify email and text messages as Spam or Not Spam (Ham).
- Implemented text cleaning, unnecessary-character removal, stemming, TF-IDF vectorization, and class-balancing techniques.
- Evaluated classification performance using Accuracy, Precision, and Recall, with emphasis on detecting unwanted messages while reducing false classifications.
- Deployed the application using Streamlit.

Email Generator

Tech: LangChain, Google Gemini, Generative AI

GitHub: github.com/Akchiche-Mohamed-Aymen/Email-generator | Demo: generate-email-with-ai.streamlit.app/

- Developed a Generative AI application that transforms user requirements into personalized professional emails.
- Designed prompts around sender information, purpose, audience, writing style, tone, emotion, and key points to generate a relevant subject line and complete email body.
- Used LangChain to orchestrate the application workflow and the Gemini API as the underlying language model.

Customer Churn Prediction & Segmentation

Tech: Python, Scikit-learn, Pandas, LangChain, Streamlit, Generative AI

GitHub: github.com/Akchiche-Mohamed-Aymen/Customer-Churn-Prediction-Segmentation-Using-Machine-Learning | Demo: customer-churn-prediction-segmentation-using-machine-learning.streamlit.app/

- Developed an end-to-end banking analytics application for customer churn prediction, customer segmentation, and AI-generated retention recommendations.
- Performed data preprocessing, exploratory analysis, feature engineering, and model evaluation to prepare customer data for predictive analytics.
- Integrated LangChain and the Gemini API to generate personalized recommendations and retention strategies based on customer risk and segment information.
- Built and deployed an interactive Streamlit application.

ROP Prediction System

Tech: Python, Scikit-learn, Pandas, NumPy, Streamlit

GitHub: github.com/Akchiche-Mohamed-Aymen/rop_prediction | Demo: rop-prediction90.streamlit.app/

- Developed a freelance predictive analytics solution for the oil and gas industry to estimate Rate of Penetration (ROP) from drilling and geological data.
- Performed data preprocessing, exploratory analysis, feature engineering, and feature selection on operational and formation-related data.
- Evaluated predictive performance using R^2 Score, MAE, and RMSE and integrated the final solution into an interactive Streamlit application.

House Price Estimator

Tech: Python, NumPy, Pandas, Streamlit

GitHub: github.com/Akchiche-Mohamed-Aymen/house-prediction-multiple-regression | Demo: house-prediction-multiple-regression-mphxy4in5viwbbujeff7oy.streamlit.app/

- Implemented a house price prediction system from scratch using Python and NumPy to strengthen understanding of predictive modeling and optimization.
- Built the data preprocessing, feature selection, optimization, and evaluation workflow without relying on high-level machine learning frameworks.
- Deployed the estimator as an interactive Streamlit application.

LEADERSHIP & INSTRUCTIONAL EXPERIENCE

Course Designer, Organizer & Instructor - Programming Fundamentals

Scientific Corner Scientific Club

- Designed and delivered a Programming Fundamentals course for university members from diverse academic backgrounds, including learners with no prior computer science experience.
- Assessed participants' technical backgrounds and translated programming and software engineering concepts into accessible learning paths.
- Developed curriculum materials combining theoretical concepts with practical exercises to strengthen problem-solving skills.
- Mentored learners across different proficiency levels and demonstrated strong technical communication and instructional skills.

LANGUAGES

- **Arabic:** Native proficiency
- **English:** B1 Professional technical proficiency